MIL-H-81581/4A
10 December 1982
SUPERSEDING
MIL-H-81581/4(AS)
8 December 1969

MILITARY SPECIFICATION SHEET

HOSE ASSEMBLIES, BREATHING OXYGEN, LOW PRESSURE, WITH/WITHOUT CONNECTIONS FOR AN INTEGRATED COMMUNICATION SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the hose assemblies described herein shall consist of this document and the issue in effect of Specification MIL-H-81581.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for two types of low pressure oxygen hose assemblies which extend from the oxygen mask to a connector or chest-mounted regulator.

1.2 Classification. Hose and hose assemblies covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Hose assembly without integrated communication cable

Type II - Hose assembly with integrated communication cable

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-81581	-	Hose assemblies, Breathing Oxygen and Air: General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

280AS100 - Hose Assembly Index, Breathing Oxygen, Low Pressure-With/ Without Connections for an Integrated Communication System.

(Copies of specifications, standards, drawings, and publications required by manufacturer's in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements should be in accordance with the general specification MIL-H-81581 and as specified herein.

3.2 Qualification. The hoses furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.1.1). Qualification will be in accordance with groups as listed on Drawing 280AS100. Qualification approval of any individual hose listed within a group will automatically qualify a manufacturer for all other items listed within the group, provided the conditions set forth in 6.3.1 of MIL-H-81581 are complied with.

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.1.2).

3.4 Design and construction. The hose assembly shall be in accordance with the general specification (except that the bore need not be smooth) and with the applicable drawing listed on Drawing 280AS100.

3.5 Performance. Unless otherwise specified herein, the performance of the hose assembly shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.5.1 Polarity (color code) and electrical continuity (Type II only). When the hose assembly is tested as specified in the general specification, the resistance between all points designed to be electrically connected, shall be as specified in the general specification, with an allowable resistance of 0.03 ohms for each lineal foot of electrical wire in the hose.

3.5.2 Elongation. When tested as specified in the general specification, the increase in length of the hose shall not exceed 50 percent for Type I and 15 percent for Type II. The permanent set shall not exceed five percent of the original hose length, when determined 2 minutes after removal of the load.

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3.5.3 Strength.

3.5.3.1 Tensile load. When tested as specified in the general specification, there shall be no indication of damage to any portion of the hose assembly as a result of this test.

3.5.3.2 Static load. When tested as specified in the general specification, the hose shall not show a deflection of more than 25 percent of original value on the outside diameter. With the load removed, the outside diameter of the hose shall be within 5 percent of original value.

3.5.4 Restraint cord elongation and tensile strength. When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds, when tested as specified in the general specification.

3.5.5 Weight. When tested as specified in the general specification, the weight of the hose assembly shall not exceed the following:

Type	Maximum weight in ounces
I	3.0
II	9.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspections. Sampling and inspections shall be in accordance with the general specification and as specified herein.

4.1.1 Qualification inspection. The qualification inspection of the hose assembly shall consist of the following examinations and tests from the general specification:

Visual examination
Polarity (color code) and electrical continuity, Procedure I (Type II only)
Insulation resistance, Procedure I (Type II only)
Elongation, Procedure II
Delamination, Procedure I
Odor, Procedure I
Cleanliness, Procedure I and II
Leakage, Procedure II
Flexibility, Procedure IV (for environmental conditions, Procedure II)
Minimum burst pressure, Procedure II
Tensile load, Procedure V
Restraint cord elongation and tensile strength
Static load, Procedure I
Low temperature, Procedure I
High temperature, Procedure I
Flexibility endurance, Procedure II
Ozone resistance, Procedure II
Abrasion, Procedure I
Weight, Procedure I

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4.1.2 First article inspection. The first article inspection of the hose shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.1.2.1 First article samples. The first article sample shall consist of four hose assemblies of the part number contracted for.

4.1.3 Quality conformance inspection. The quality conformance inspection shall consist of the following tests from those required by the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Polarity (color code) and electrical continuity, Procedure I (Type II only)
Insulation resistance, Procedure I (Type II only)
Elongation, Procedure II
Delamination, Procedure I
Odor, Procedure I
Cleanliness, Procedure I and II
Leakage, Procedure II
Flexibility, Procedure IV
Packaging

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

6.1 Intended Use. The hose assemblies covered by this detail specification are intended for use with the oxygen mask and a connector or chest-mounted regulator in supplying breathing oxygen to aircrewman in the operation of aircraft.

6.2 Ordering data. Ordering data shall be in accordance with the general specification.

Custodians:
Army - AV
Navy- AS
Air Force - 11

Review activity:
Air Force - 99

Preparing activity:
Navy - AS

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